

Metro Ticket Generating System

PROJECT DESIGN PHASE

Date	27 July 2026
Name	T V SAI LAXMI
Project Name	Metro Ticket Generating System
Maximum Marks	4 Marks

1. Problem-Solution Fit

1.1 Problem Recap

Metro commuters continue to depend on manual ticket counters, cash-based token machines, and partially digital kiosks to travel. This forces daily commuters, students, senior citizens, and differently-abled passengers to stand in long queues, calculate fares manually, and rely on physical tokens or paper tickets that are easy to misplace. During peak office hours, exam seasons, and festival rush, this manual dependency leads to delays, overcrowding at counters, and an overall inconvenient and inaccessible travel experience.

1.2 Problem-Solution Fit Statement

"Because metro ticket booking today is manual, queue-dependent, and inaccessible to many commuters, a digital self-service booking system built on ServiceNow — with automated fare calculation and instant QR-code ticket delivery — directly removes the counter dependency and gives every commuter a fast, reliable, and accessible way to travel."

1.3 Problem-to-Solution Mapping

Commuter Pain Point	Root Cause	How the Proposed Solution Solves It
Long queues at ticket counters during peak hours	Limited counter staff and manual, one-at-a-time processing	Self-service Service Catalog booking form accessible anytime from any device, removing the need to queue
Manual fare calculation errors	Fare charts are checked and applied by hand	Flow Designer automatically computes fare based on source, destination, and passenger count
Limited counter operating hours vs. 24x7 travel needs	Physical counters follow fixed staff shift timings	Digital booking portal available round the clock, independent of station staff hours
Lost, damaged, or forgotten physical tokens	Tickets exist only as a physical token or paper slip	Digital QR-code ticket stored and delivered instantly via notification, no physical item to lose
Poor support for elderly and differently-abled commuters	Counters offer inconsistent, staff-dependent assistance	Consistent, guided, step-by-step digital booking flow usable independently by every commuter
No visibility into fare or booking history	No digital trail of past manual purchases	Booking history and status tracked automatically against every Service Catalog request record

1.4 Why This Solution Fits

- Solves the core commuter frustration — time lost to manual, in-person processes — by moving the entire booking journey onto a self-service digital platform.
- Removes fare uncertainty by calculating and displaying the exact fare before payment is made.
- Matches commuter expectations of booking a ticket "from the phone, like booking a cab," identified directly in the empathy mapping exercise.
- Uses only core ServiceNow capabilities — Service Catalog, Flow Designer, and Notifications — making the solution realistic to build and maintain.
- Addresses accessibility gaps by offering a single, consistent booking experience for every category of commuter.

1.5 Fit Validation by Stakeholder

Commuters

- Book a ticket in under a minute, from anywhere
- No time lost standing in queues
- Fare visibility before travel

Metro Authority

- Lower operational cost at counters
- Reduced revenue leakage from manual fare errors
- Less counter congestion during peak hours

Elderly / Differently-Abled

- Independent, self-service booking
- No reliance on counter staff assistance
- Consistent experience every time

City / Environment

- Higher metro adoption over private vehicles
- Smoother, faster commuter movement
- Support for sustainable public transport use

1.6 Conclusion

The problem-solution fit clearly demonstrates that the primary drivers of commuter frustration—manual queues, cash handling, physical token loss, and human error—are directly mitigated by transitioning to a digital, self-service ticketing system built on ServiceNow.

By unifying automated fare calculation via Flow Designer, instant QR-code generation, and real-time status tracking into a single, intuitive Service Catalog interface, the proposed solution accomplishes three critical objectives:

1. **Enhances Commuter Experience:** Eliminates counter dependency, reduces ticket purchase times to under a minute, and ensures seamless accessibility across all demographic groups.
2. **Optimizes Station Operations:** Reduces peak-hour congestion at physical counters, eliminates manual fare discrepancies, and lowers administrative overhead for the Metro Authority.
3. **Ensures Scalability & Reliability:** Leverages ServiceNow's enterprise-grade workflow engine to provide round-the-clock availability with zero reliance on station staff operating hours.

In summary, this ServiceNow-based digital framework successfully aligns commuter needs with operational efficiency. It validates moving forward into the execution phase with **Service Catalog integration**, **Flow Designer automation**, and **instant notification delivery** as the core foundation of the Metro Ticket Generating System.